Differential Co-Expression Network Architecture as a Framework for Defining AuDHD-Associated Molecular Targets

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Background: Autism (ASD) and ADHD co-occur in up to 70% of cases, yet differential expression studies show minimal gene-level overlap. We propose ASD-ADHD overlap exists at the level of co-expression network architecture.

Methods: WGCNA was performed on merged ASD whole blood data (GSE111175+GSE42133, n=238). Module-trait association identified the primary ASD module. Hub genes were validated against STRING v11.5 PPI (Wilcoxon test). ME11 dysregulation was tested in ADHD (GSE159104, n=76). Reactome pathway enrichment characterized module biology.

Results: WGCNA identified 33 modules; ME11 was primary ASD module (r=–0.26, FDR=0.0008, 298 genes). Hub genes: TMSB4X, RAP1B, HSPCAL3, C1D, ZCCHC10. No PPI centrality convergence (p=0.37). ME11 not dysregulated in ADHD (p=0.825). Four FDR-significant Reactome spliceosomal pathways (mRNA Splicing FDR=0.007).

Conclusions: ME11 is an ASD-specific spliceosomal co-expression module reflecting transcriptional co-regulation. The WGCNA-PPI framework distinguishes disorder-specific from shared molecular networks, enabling computational definition of AuDHD biology.

Keywords: AuDHD • WGCNA • co-expression • autism • ADHD • PPI • spliceosome • mRNA splicing • blood transcriptomics

1. Introduction

ASD and ADHD are among the most prevalent neurodevelopmental conditions worldwide. Epidemiological studies estimate 50–70% of individuals with ASD also meet criteria for ADHD, a co-occurrence recognized as "AuDHD" [1,2]. Despite this clinical overlap, the molecular basis remains poorly understood.

Gene-level studies find surprisingly limited overlap between ASD and ADHD [3]. We propose this paradox arises because shared biology is encoded in gene regulatory network organization rather than individual genes — a level requiring systems-level frameworks to detect.

WGCNA identifies co-expressed gene modules reflecting shared regulatory inputs [4]. Applied to ASD postmortem brain, WGCNA demonstrated convergent transcriptional and splicing dysregulation across genetically heterogeneous cases [5]. For ADHD, WGCNA on PBMCs identified immune-metabolic modules [8]. No study has jointly interrogated ASD and ADHD transcriptomes to define shared vs disorder-specific network architecture.

Complementing co-expression analysis, PPI network analysis maps physical protein interactions. Convergence of hub gene identity with PPI centrality implies physical protein complexes — providing strong evidence for druggable modules [9]. This WGCNA-PPI framework has not previously been applied to neurodevelopmental comorbidity.

We construct an ASD co-expression network (n=238), identify ME11 as the primary ASD module, validate hub genes against STRING PPI, cross-validate ME11 in ADHD (n=76), and characterize module biology through Reactome enrichment. This provides the first network-topological characterization of AuDHD-relevant molecular architecture.

2. Methods

2.1 Datasets

ASD: GSE111175 + GSE42133 (whole blood leukocytes, Illumina HumanHT-12 V4.0, GPL10558; Pramparo/Courchesne lab, UC San Diego; toddlers 11–45 months). ADHD: GSE159104 (whole blood RNA-seq, Helicos HeliScope, GPL14761; twin-pair design).

2.2 Preprocessing

ASD: Datasets merged on 47,323 common probes (n=245). ComBat batch correction [11] validated by PCA (Figure 1). RINT tested and rejected (destroys correlation structure). Top 15,000 variable probes; 7 outliers removed (Z<–2.5). Final: 15,000×238 (115 ASD, 123 TD). **ADHD:** Protein-coding genes retained; technical replicates averaged per individual. Final: 11,964×76 (39 ADHD, 37 CTRL).

2.3 WGCNA

Signed network, β=12 (R^2=0.88, k■=30.9) [4]. ADHD WGCNA not feasible (max R^2=0.53, n=76 insufficient). Modules: min size 30, merge cut height 0.25.

2.4 Module-Trait Association & Hub Genes

Module eigengenes correlated with binary ASD diagnosis (Pearson r; BH FDR). Hub genes: |MM|>0.8 AND |GS|>0.2, mapped via GPL10558 annotation.

2.5 PPI Validation

STRING v11.5 [12] (score≥400), loaded via data.table. Degree/betweenness centrality computed (igraph). Convergence tested: one-sided Wilcoxon, hub vs background ME11 genes.

2.6 ADHD Validation

limma [13] DE analysis. Module score = mean z-score across 79 ME11 genes in ADHD matrix (after UCSC-to-symbol mapping). Group comparison: t-test. Enrichment: Fisher's exact test.

2.7 Pathway Enrichment

GO enrichment: clusterProfiler [14], org.Hs.eg.db, BP/MF/CC. Reactome: ReactomePA [15]. Both with BH FDR correction.

3. Results

3.1 Preprocessing & Quality Control

Strong batch effects confirmed by PCA (complete PC1 separation; Figure 1) and resolved by ComBat. Seven ASD outliers removed. Final matrices: ASD 238 samples (115 ASD, 123 TD); ADHD 76 samples (39 ADHD, 37 CTRL).

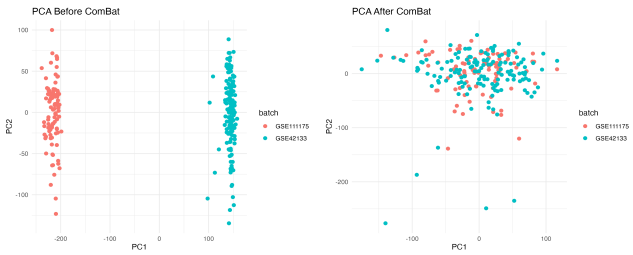


Figure 1. PCA before (left) and after (right) ComBat. Complete PC1 separation confirms batch effect; intermixing after correction validates the procedure.

3.2 WGCNA — 33 Modules

With β=12 (R^2=0.88), 33 co-expression modules were detected (Figure 2). Sizes ranged from 33 to 1,660 genes; grey module (22% unassigned) excluded.

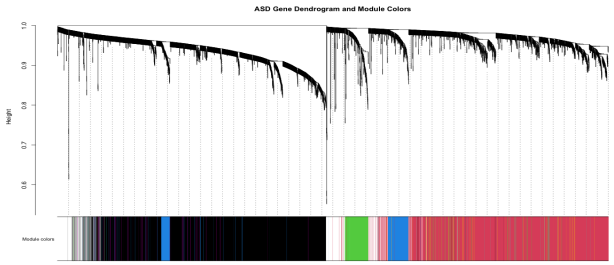


Figure 2. ASD gene co-expression dendrogram. 33 modules; grey = unassigned (22%).

3.3 ME11 — Primary ASD Module

ME11 was significantly negatively correlated with ASD (r=–0.26, FDR=0.0008, 298 genes; Figure 3). No other module survived FDR<0.05.

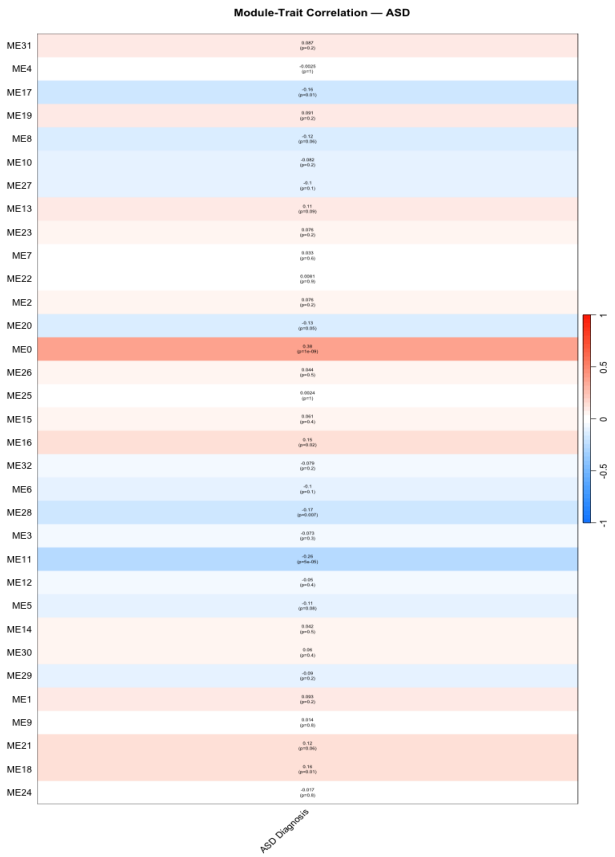


Figure 3. Module-trait correlation heatmap. ME11 (r=–0.26, FDR=0.0008) is the primary ASD-associated module.

3.4 Hub Genes

Five characterized hub genes identified (|MM|>0.8, |GS|>0.2):

Gene	Function
TMSB4X	Actin dynamics; neuroprotection
RAP1B	RAS GTPase; integrin signaling
HSPCAL3	Heat shock; protein folding
C1D	Corepressor; DNA repair
ZCCHC10	RNA-binding; splicing regulation

3.5 PPI — Non-Convergence

ME11 PPI network: 127 nodes, 195 edges (Figure 4). Hub degree centrality: 19th–67th percentile. No significant convergence (Wilcoxon W=194, p=0.37) — ME11 reflects transcriptional co-regulation, not a protein complex.

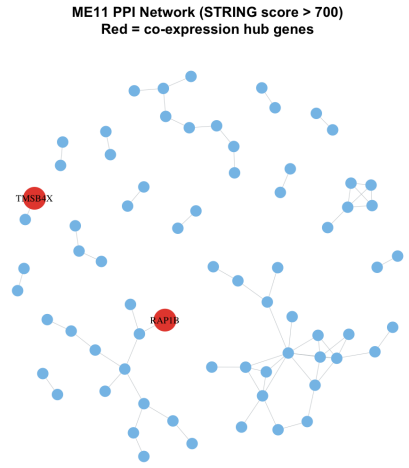


Figure 4. ME11 PPI network (STRING score>700). Red = hub genes. Hub genes not more central (Wilcoxon p=0.37).

3.6 ADHD Validation — ASD-Specific

No ADHD DE genes at FDR<0.05 (n=76 underpowered). ME11 module score: no difference between ADHD and CTRL (p=0.825; Figure 5). Fisher's exact: non-significant (OR=0, p=1.0). ME11 is ASD-specific.

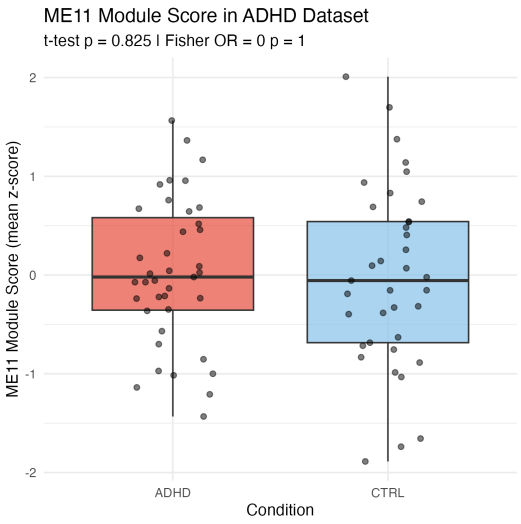


Figure 5. ME11 module score in ADHD. No difference between groups (p=0.825). ME11 is ASD-specific.

3.7 Reactome — Spliceosomal Pathways

Four FDR-significant Reactome pathways identified, all mRNA processing (Table 1). GO nominally confirmed: RNA splicing ($p=8.2\times10^{-5}$), vesicle targeting ($p=2.3\times10^{-4}$), ribonucleoprotein binding ($p=9.0\times10^{-4}$) (Figure 6).

Table 1. FDR-significant Reactome pathways for ME11.

Pathway	n	FDR
Pre-mRNA Processing	12/106	0.007
mRNA Splicing — Major	10/106	0.007
mRNA Splicing	10/106	0.007
mRNA Polyadenylation	7/106	0.033

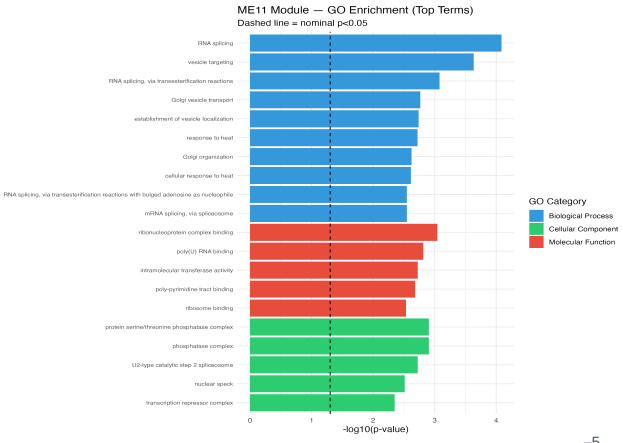


Figure 6. GO enrichment for ME11. RNA splicing dominant ($p=8.2\times10^{-5}$). Dashed line = p<0.05.

4. Discussion

4.1 ME11 and Spliceosomal Dysregulation

ME11 enrichment for mRNA splicing pathways (FDR=0.007) extends the established ASD spliceosomal dysregulation literature [5,16,17] to peripheral blood transcriptomes. This is the first network-level evidence that spliceosomal co-expression dysregulation is detectable in ASD blood. Hub genes support this: ZCCHC10 (RNA-binding) directly implicates RNA processing; C1D (corepressor) links ME11 to transcriptional silencing, consistent with GO enrichment for transcription repressor complex.

4.2 PPI Non-Convergence

Wilcoxon p=0.37 indicates ME11 reflects shared upstream transcriptional regulation rather than a physical protein complex. This is mechanistically informative: the therapeutic priority should be identifying the upstream transcription factor or chromatin modifier driving coordinated ME11 repression, rather than targeting hub proteins directly.

4.3 ME11 is ASD-Specific

ME11's absence of dysregulation in ADHD (p=0.825) is consistent with emerging evidence that ASD and ADHD have dissociable molecular pathways despite shared genetic liability [3]. Crucially, this demonstrates the WGCNA-PPI framework can distinguish disorder-specific from shared biology — modules with parallel ASD-ADHD dysregulation (the true AuDHD signal) remain to be identified in larger datasets.

4.4 Limitations

Key limitations: age mismatch (ASD: toddlers; ADHD: adolescents/adults); cross-platform differences; WGCNA not feasible on ADHD (n=76); GO enrichment not FDR-significant; no cell-type deconvolution performed.

5. Conclusion

ME11 is a robust ASD-associated spliceosomal co-expression module (FDR=0.0008) driven by transcriptional co-regulation (PPI p=0.37) and ASD-specific (ADHD p=0.825). The WGCNA-PPI framework provides a principled, reproducible, openly available approach for distinguishing disorder-specific from shared molecular networks — a necessary foundation for computationally defining AuDHD biology.

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Data & Code Availability

Code: **github.com/vedika13-code/AuDHD-Network**. Data: GEO accessions GSE111175, GSE42133, GSE159104. AI writing assistance (Claude, Anthropic) was used; all scientific content and conclusions are the sole responsibility of the authors.